COMMERCIAL ITEM DESCRIPTION

WIRE, NONELECTRICAL, (Platinum Alloy)

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-W-16193B.

Salient characteristics:

This commercial item description covers platinum alloy wire for use in direct casting to invested parts of dental prosthetic appliances.

The platinum alloy wire shall be of the following composition:

Gold - 20.0 - 20.5 percent.
Platinum - 50.0 - 50.5 percent.
Palladium - 26.0 - 26.5 percent.
Silver - Remainder.

The platinum alloy wire shall be suitable for use in fabricating wrought prosthodontic clasps, pins, and posts to which precious metal dental alloys can be cast.

Two straight wires shall be furnished for each unit of issue. Each wire shall measure 9 inches, minimum, in length and shall be 0.0400 inch ± 0.0005 inch in diameter.

Individual units of issue of platinum alloy wire shall weigh 6.220 grams ± 0.062 grams. The total weight of wire for each delivery to the Government shall have no minus weight tolerance.

Physical properties.

Ultimate tensile strength. Ultimate tensile strength of the wire shall be not less than 120,000 pounds per square inch when given the heat treatment as specified below.

Elongation. The elongation of the wire shall be not less than 10 percent as received from the manufacturer, and not less than 10 percent when given the heat treatment as specified below.

Finish. The wire shall have a bright, smooth finish.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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Instructions. Adequate instructions for heat treating and using the platinum alloy wire shall be furnished with each package.

Identification marking. Each length of wire shall be die stamped with an identification or trademark at one end. Ends may be flattened for this purpose.

Workmanship. The platinum alloy wire shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). Each four pennyweight of wire in two lengths, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties). Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government.

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Sampling for examination. The inspection levels and acceptable quality levels (AQL's) shall be as listed below. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5, and 4.0 percent defective, and an inspection level of II. The unit of product for examination shall be one unit of issue.

Examination. Wire shall be examined for defects including, but not limited to, those listed below.

Defects.

Diameter of wire not as specified.
Workmanship defects such as cracks, kinks, laminations and pitted areas present.
Finish not as specified.
Units of issue of wire not furnished in 2 lengths.
Identification marking not as specified.
Instructions missing, incomplete, illegible, or incorrect.
Length of wire not as specified.

Testing. A sufficient quantity of platinum alloy wire of each lot shall be selected for the chemical analysis test. Five units of issue of wire from each lot shall be tested for individual weight and the entire quantity of wire delivered to the Government shall be weighed. No failures shall be allowed in these tests. The average of the results obtained shall be used for determining compliance with tensile strength and elongation requirements.

Weight. Weights of individual units of issue of platinum alloy wire shall be determined using a suitable scale or balance accurate to 0.001 gram. The total weight of each delivery of wire to the Government shall be determined using a scale or balance accurate to 0.01 gram.

Chemical analysis. The platinum alloy wire shall be analyzed in accordance with method 111 or 112 of Federal Standard No. 151.

Specimen preparation. Specimens of platinum alloy wire for ultimate tensile strength and elongation tests shall be given the heat treatments specified below before testing.

Softening heat treatment. Selected specimens of wire shall be placed in a furnace at 1652°F (900°C) for 10 minutes and quenched in water at room temperature.

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Hardening heat treatment. Specimens of wire which have previously been given the softening heat treatment above shall be placed in a furnace at 840°F (449°C). After being held at 840°F for 2 minutes, the temperature shall be gradually reduced to 480°F (249°C) in 30 minutes (12°F per minute). Then the specimens shall be quenched in water at room temperature.

Ultimate tensile strength. Determination of ultimate tensile strength shall be made with a suitable breaking strength tester on hardened specimens of wire. The gage length shall be 2 inches. The load shall be applied slowly and continuously at a rate of 1 inch per minute until rupture occurs. The test shall be run on at least 5 specimens from each lot of wire. Report the average of the results.

Elongation. Determination of elongation of the wire shall be made on: (a) specimens as received from the contractor, (b) softened specimens, and (c) hardened specimens with a suitable breaking strength tester. The gage length shall be 2 inches. The load shall be applied slowly and continuously at a rate of 1 inch per minute until rupture occurs. Elongation observed shall be recorded to the nearest 0.5 percent. Specimens which break outside the gage marks may be repulled. The average result for 5 specimens from each lot of wire (5 each of conditions (a), (b), and (c) above) shall be reported.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

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Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

National Stock Number	Item Identification
6520-00-580-6950	WIRE, NONELECTRICAL, Platinum Alloy, Round, 4 Pennyweight, 18 Gage

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA - DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2085

Location: NCR Disk#6

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53490	HUMIDIFIER, HYGROSCOPIC

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street City State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *Last First MI* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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